Final Project - Part1

Dan Nguyen

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1 - Read Data and EDA

```r
# Read Data
bike_raw <- read.csv("bike_sharing_daily.csv")

# Re-assign values of variables
bike_raw <- bike_raw %>%
  mutate(
    season     = factor(season,     levels = 1:4,
                        labels = c("Spring", "Summer", "Fall", "Winter")),
    mnth       = factor(mnth,       levels = 1:12),
    holiday    = factor(holiday,    levels = c(0,1), labels = c("No", "Yes")),
    weekday    = factor(weekday,    levels = 0:6,
                        labels = c("Sun","Mon","Tue","Wed","Thu","Fri","Sat")),
    workingday = factor(workingday, levels = c(0,1), labels = c("No", "Yes")),
    weathersit = factor(weathersit, levels = 1:4,
                        labels = c("Clear", "Mist/Cloudy", "Light Rain/Snow", "Heavy Rain/Snow"))
  )

# Filter for 2011
bike_2011 <- bike_raw %>% filter(yr == 0)
```

Categorical Variables

Season, month, holiday, weekday, and workingday are standard characteristics within a year. There is no reason to expect significant variation in their distributions. Thus, we will not print out summary tables and plots for these variables.

Weather Situation

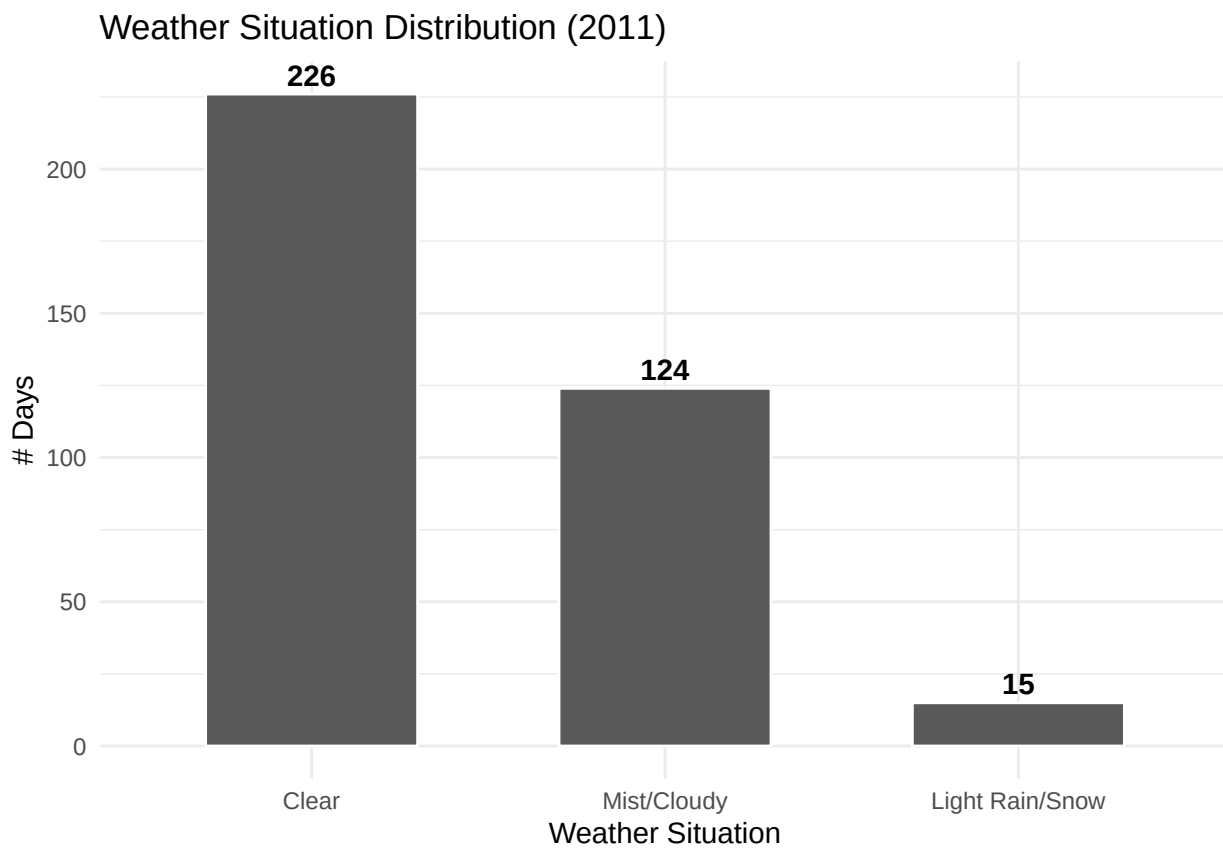
```r
table(bike_2011$weathersit)
```

```
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##             226             124              15               0
```

```r
prop.table(table(bike_2011$weathersit))
```

```
##
##           Clear     Mist/Cloudy Light Rain/Snow Heavy Rain/Snow
##      0.61917808      0.33972603      0.04109589      0.00000000
```

```r
bike_2011 %>%
  count(weathersit) %>%
  ggplot(aes(x = weathersit, y = n)) +
  geom_col(width = 0.6, color = "white") +
  geom_text(aes(label = n), vjust = -0.4, fontface = "bold") +
  labs(title = "Weather Situation Distribution (2011)",
       x = "Weather Situation", y = "# Days") +
  theme_minimal()
```



Continuous Variables

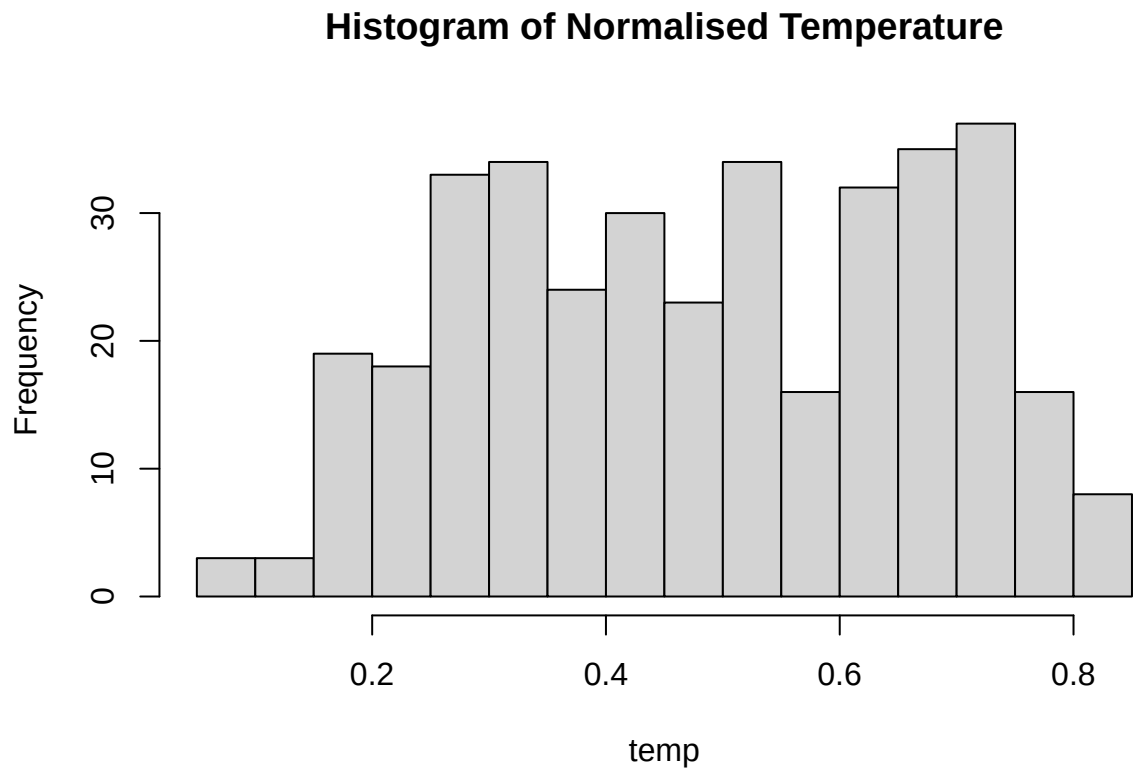
```r
cont_vars <- bike_2011 %>% select(temp, atemp, hum, windspeed, casual, registered, cnt)

# Numerical summary for all continuous variables
summary(cont_vars)
```

```
##       temp             atemp              hum           windspeed
##  Min.   :0.05913   Min.   :0.07907   Min.   :0.0000   Min.   :0.02239
##  1st Qu.:0.32500   1st Qu.:0.32195   1st Qu.:0.5383   1st Qu.:0.13558
##  Median :0.47917   Median :0.47285   Median :0.6475   Median :0.18690
##  Mean   :0.48666   Mean   :0.46684   Mean   :0.6437   Mean   :0.19140
##  3rd Qu.:0.65667   3rd Qu.:0.61238   3rd Qu.:0.7421   3rd Qu.:0.23508
##  Max.   :0.84917   Max.   :0.84090   Max.   :0.9725   Max.   :0.50746
##      casual         registered        cnt
##  Min.   :   9.0   Min.   : 416   Min.   : 431
##  1st Qu.: 222.0   1st Qu.:1730   1st Qu.:2132
##  Median : 614.0   Median :2915   Median :3740
```

```
##  Mean   : 677.4   Mean   :2728   Mean   :3406
##  3rd Qu.: 871.0   3rd Qu.:3632   3rd Qu.:4586
##  Max.   :3065.0   Max.   :4614   Max.   :6043
```

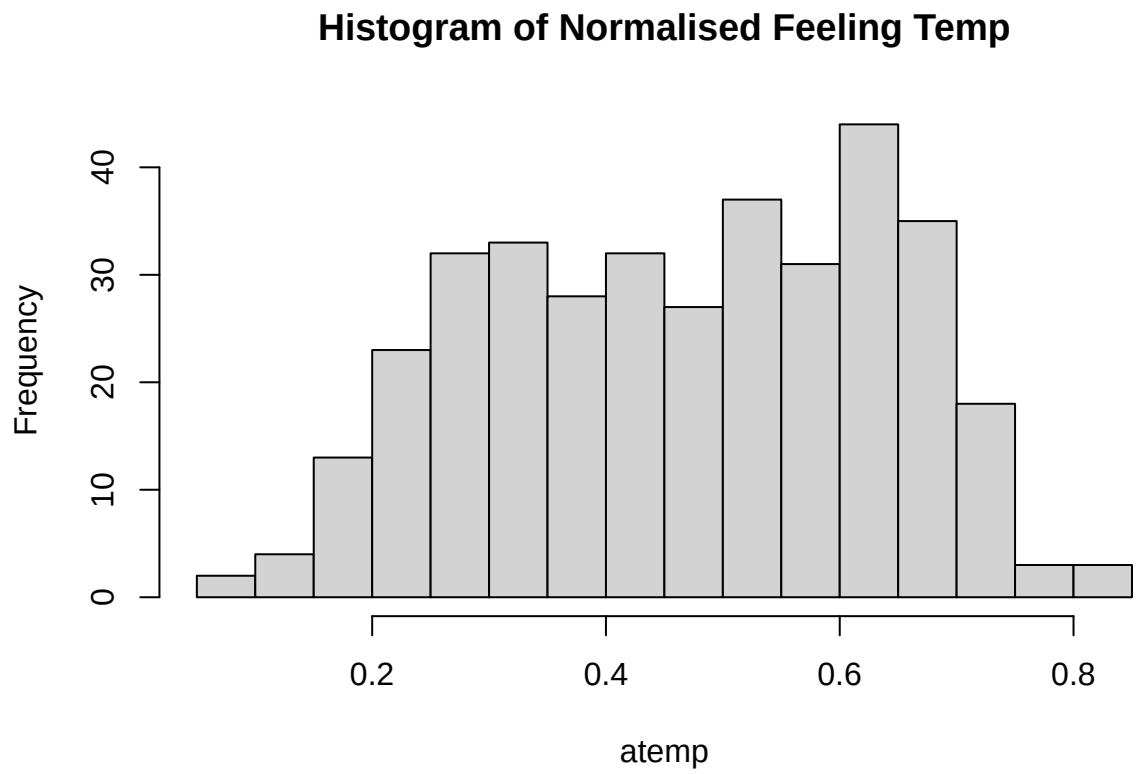
Temperature (`temp`)

```r
hist(bike_2011$temp, main = "Histogram of Normalised Temperature",
     xlab = "temp", breaks = 20)
```



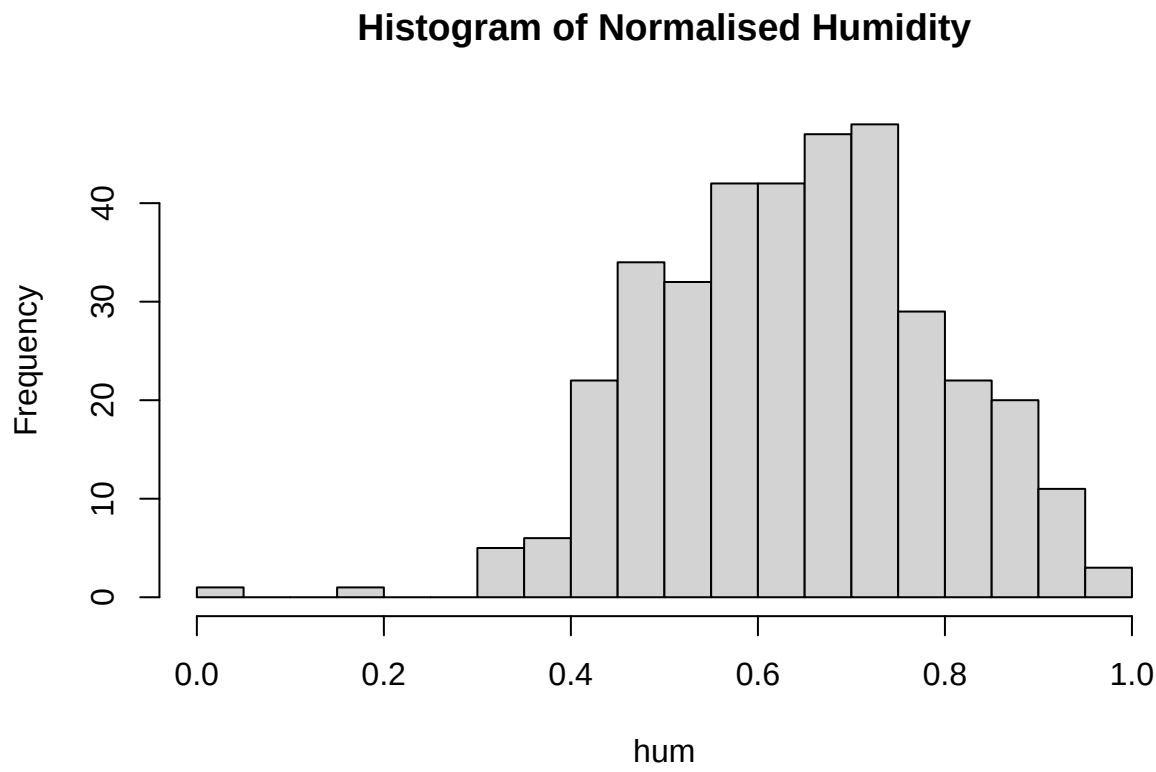
Feeling Temperature (`atemp`)

```r
hist(bike_2011$atemp, main = "Histogram of Normalised Feeling Temp",
     xlab = "atemp", breaks = 20)
```



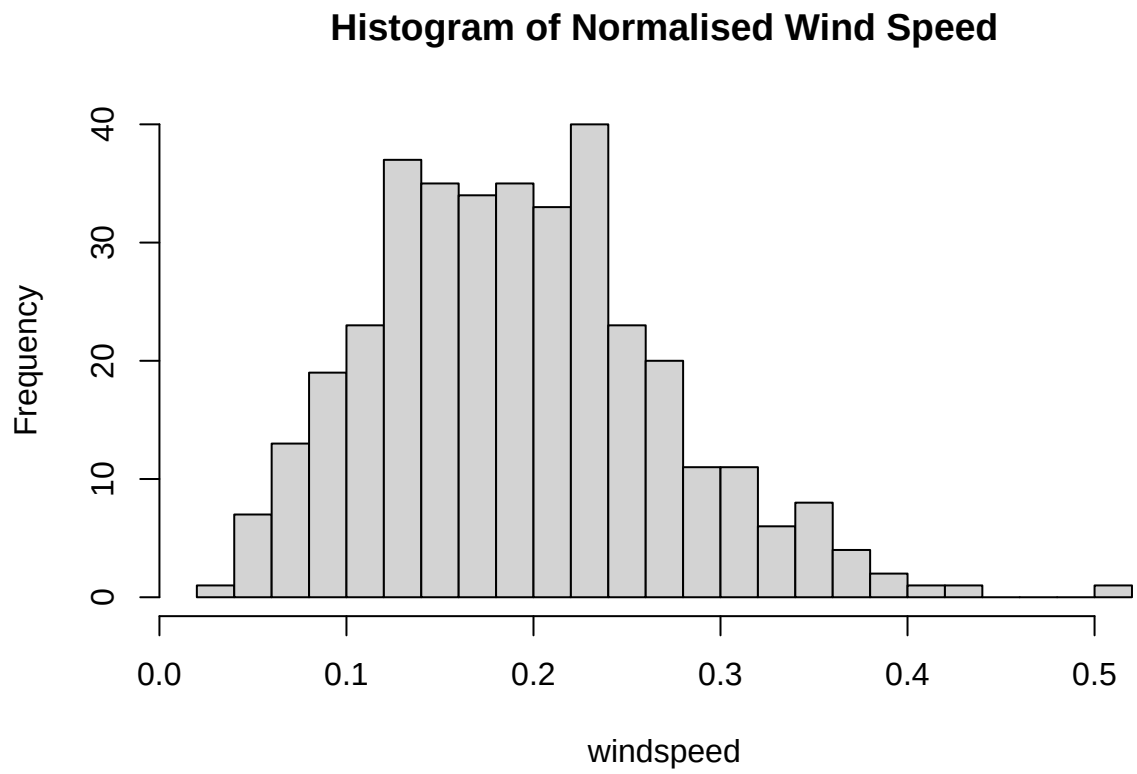
Humidity (`hum`)

```
hist(bike_2011$hum, main = "Histogram of Normalised Humidity",
     xlab = "hum", breaks = 20)
```



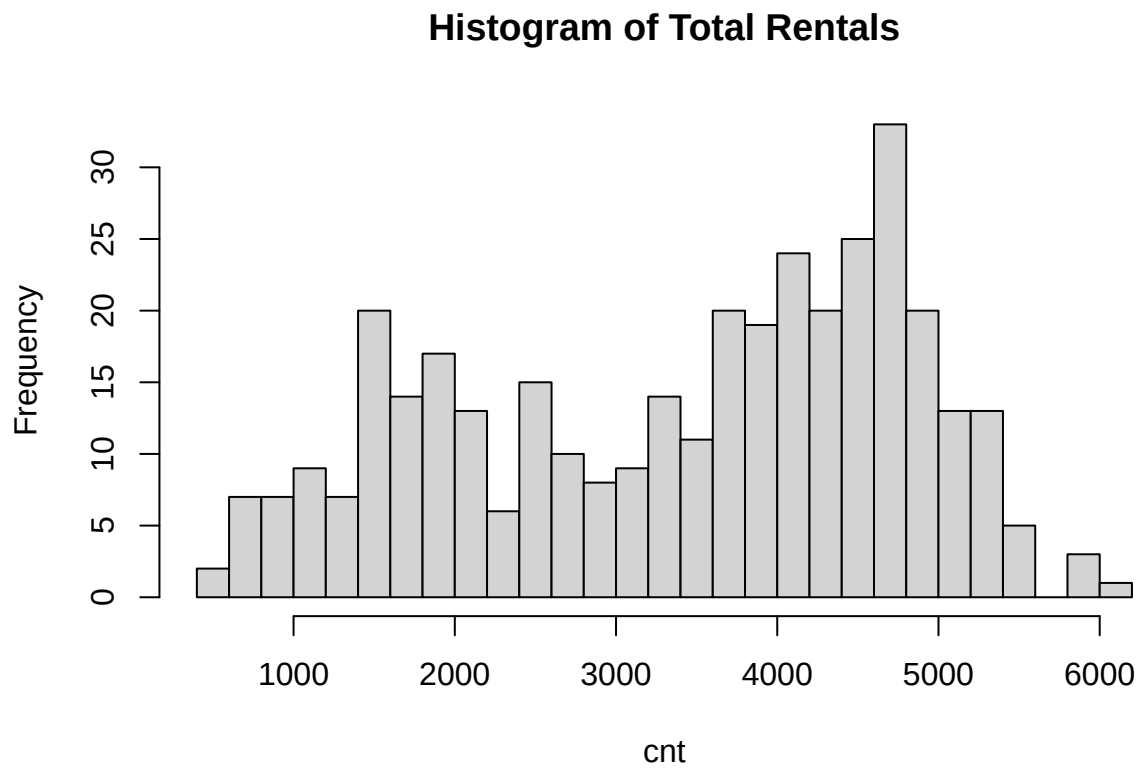
Wind Speed (`windspeed`)

```
hist(bike_2011$windspeed, main = "Histogram of Normalised Wind Speed",
     xlab = "windspeed", breaks = 20)
```



Total Rentals (`cnt`)

```
hist(bike_2011$cnt, main = "Histogram of Total Rentals",
     xlab = "cnt", breaks = 25)
```



2 - Fit linear regression models predicting `cnt` using each of the variables separately.

cnt ~ season

```r
model_season <- lm(cnt ~ season, data = bike_2011)
summary(model_season)
```

```
##
## Call:
## lm(formula = cnt ~ season, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3349.4  -504.7    96.8   593.6  2029.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1666.67      95.43   17.46   <2e-16 ***
## seasonSummer  2108.51     134.23   15.71   <2e-16 ***
## seasonFall    2797.70     133.52   20.95   <2e-16 ***
## seasonWinter  1997.79     135.34   14.76   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 905.4 on 361 degrees of freedom
## Multiple R-squared:  0.5724, Adjusted R-squared:  0.5688
## F-statistic: 161.1 on 3 and 361 DF,  p-value: < 2.2e-16
```

The model explains about 57% of the variance in total rentals. The coefficients indicate that, compared to Spring, summer has 2108 more rentals, fall has 2797 more rentals, and winter has 1997 more rentals on average. Overall, season is a good predictor of total rentals.

cnt ~ mnth

```r
model_mnth <- lm(cnt ~ mnth, data = bike_2011)
summary(model_mnth)
```

```
##
## Call:
## lm(formula = cnt ~ mnth, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3357.2  -315.2    66.0   479.7  2149.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1231.9      137.6   8.950  < 2e-16 ***
## mnth2          490.1      199.8   2.453   0.0147 *
## mnth3          834.1      194.7   4.285 2.36e-05 ***
## mnth4         1930.4      196.3   9.835  < 2e-16 ***
## mnth5         3149.4      194.7  16.179  < 2e-16 ***
## mnth6         3551.8      196.3  18.096  < 2e-16 ***
## mnth7         3327.5      194.7  17.094  < 2e-16 ***
## mnth8         3177.5      194.7  16.323  < 2e-16 ***
## mnth9         3015.4      196.3  15.363  < 2e-16 ***
## mnth10        2752.3      194.7  14.139  < 2e-16 ***
## mnth11        2173.7      196.3  11.075  < 2e-16 ***
## mnth12        1585.0      194.7   8.142 6.75e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 766.4 on 353 degrees of freedom
## Multiple R-squared:  0.7004, Adjusted R-squared:  0.691
## F-statistic: 75.01 on 11 and 353 DF,  p-value: < 2.2e-16
```

The model has a better R2 than season, explaining about 70% of the variance in total rentals. Due to the strong corelation between month and season, the coefficients for month show a clear seasonal pattern. Rentals peak in the summer months (6, 7, 8).

cnt ~ holiday

```r
model_holiday <- lm(cnt ~ holiday, data = bike_2011)
summary(model_holiday)
```

```
##
## Call:
## lm(formula = cnt ~ holiday, data = bike_2011)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2986.1 -1285.1   329.9  1168.9  3040.8
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3417.13      73.19  46.689   <2e-16 ***
## holidayYes   -414.93     442.17  -0.938    0.349
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1379 on 363 degrees of freedom
## Multiple R-squared:  0.00242,    Adjusted R-squared:  -0.0003282
## F-statistic: 0.8806 on 1 and 363 DF,  p-value: 0.3487
```

The model has a very low R2 of 0.002, indicating that holiday status is not a good predictor of total rentals.

cnt ~ weekday

```r
model_weekday <- lm(cnt ~ weekday, data = bike_2011)
summary(model_weekday)
```

```
##
## Call:
## lm(formula = cnt ~ weekday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2925.8 -1273.1   345.2  1147.7  2577.2
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3405.27     192.49  17.691   <2e-16 ***
## weekdayMon     60.52     272.22   0.222    0.824
## weekdayTue     62.77     272.22   0.231    0.818
## weekdayWed   -152.02     272.22  -0.558    0.577
## weekdayThu    -48.50     272.22  -0.178    0.859
## weekdayFri     94.85     272.22   0.348    0.728
## weekdaySat    -13.89     270.93  -0.051    0.959
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1388 on 358 degrees of freedom
## Multiple R-squared:  0.003175,   Adjusted R-squared:  -0.01353
## F-statistic: 0.1901 on 6 and 358 DF,  p-value: 0.9795
```

This model also has a low R2 of 0.003, suggesting that day of the week is a bad predictor of total rentals.

cnt ~ workingday

```r
model_workingday <- lm(cnt ~ workingday, data = bike_2011)
summary(model_workingday)
```

```
##
## Call:
```

```
## lm(formula = cnt ~ workingday, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2994.1 -1291.1   321.9  1164.9  2679.2
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3363.82     128.72  26.133   <2e-16 ***
## workingdayYes    61.24     155.53   0.394    0.694
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1380 on 363 degrees of freedom
## Multiple R-squared:  0.0004269,  Adjusted R-squared:  -0.002327
## F-statistic: 0.155 on 1 and 363 DF,  p-value: 0.694
```

This model's R2 is worse. At 0.0004, it indicates that whether it's a working day or not has no predictive power for total rentals.

cnt ~ weathersit

```r
model_weathersit <- lm(cnt ~ weathersit, data = bike_2011)
summary(model_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ weathersit, data = bike_2011)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##  -3264  -1103    373   1032   2955
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)                3694.99      86.74  42.598  < 2e-16 ***
## weathersitMist/Cloudy      -606.89     145.73  -4.164 3.90e-05 ***
## weathersitLight Rain/Snow -2020.85     347.69  -5.812 1.35e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1304 on 362 degrees of freedom
## Multiple R-squared:  0.1104, Adjusted R-squared:  0.1055
## F-statistic: 22.46 on 2 and 362 DF,  p-value: 6.363e-10
```

This model has an R2 of 0.11. The coefficients show that, compared to clear weather, adverse weather conditions (mist/cloudy, light rain/snow, heavy rain/snow) are associated with fewer rentals. However, the model's overall predictive power is still quite low.

cnt ~ temp

```r
model_temp <- lm(cnt ~ temp, data = bike_2011)
summary(model_temp)
```

```
##
```

```
## Call:
## lm(formula = cnt ~ temp, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3375.0  -501.2   -60.3   677.3  1985.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    676.4      126.9   5.331 1.72e-07 ***
## temp          5608.3      243.0  23.082  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 878.9 on 363 degrees of freedom
## Multiple R-squared:  0.5948, Adjusted R-squared:  0.5937
## F-statistic: 532.8 on 1 and 363 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.59, indicating that temperature is a strong predictor of total rentals. The positive coefficient suggests that higher temperatures are associated with more rentals.

cnt ~ atemp

```r
model_atemp <- lm(cnt ~ atemp, data = bike_2011)
summary(model_atemp)
```

```
##
## Call:
## lm(formula = cnt ~ atemp, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3359.0  -536.9   -41.5   680.9  1924.1
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    450.1      134.4    3.35 0.000893 ***
## atemp         6331.2      270.7   23.39  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 872 on 363 degrees of freedom
## Multiple R-squared:  0.6011, Adjusted R-squared:    0.6
## F-statistic:   547 on 1 and 363 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.6, slightly better than temp. The positive coefficient indicates that higher feeling temperatures are associated with more rentals.

cnt ~ hum

```r
model_hum <- lm(cnt ~ hum, data = bike_2011)
summary(model_hum)
```

```
##
## Call:
```

```
## lm(formula = cnt ~ hum, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2975.5 -1272.9   333.9  1180.9  2637.3
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3394.44     321.38  10.562   <2e-16 ***
## hum            17.59     486.51   0.036    0.971
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1381 on 363 degrees of freedom
## Multiple R-squared:  3.603e-06,  Adjusted R-squared:  -0.002751
## F-statistic: 0.001308 on 1 and 363 DF,  p-value: 0.9712
```

This model has a very low R2, the p-value for hum is also not significant. This suggests that humidity is not a good predictor of total rentals.

cnt ~ windspeed

```r
model_windspeed <- lm(cnt ~ windspeed, data = bike_2011)
summary(model_windspeed)
```

```
##
## Call:
## lm(formula = cnt ~ windspeed, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3361.4 -1071.1   275.1  1068.4  2601.7
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4359.9      186.4  23.384  < 2e-16 ***
## windspeed    -4985.0      904.1  -5.514 6.67e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1326 on 363 degrees of freedom
## Multiple R-squared:  0.07728,    Adjusted R-squared:  0.07474
## F-statistic:  30.4 on 1 and 363 DF,  p-value: 6.673e-08
```

This model has a low R2 pf 0.07, indicating that wind speed is a weak predictor of total rentals.

3 - Fit multiple linear regression models with interactions.

Given that season, mnth, weathersit, temp, atemp are good predictors individually, we will explore interactions between these variables.
cnt ~ temp*weathersit

```r
model_temp_weathersit <- lm(cnt ~ temp * weathersit, data = bike_2011)
summary(model_temp_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ temp * weathersit, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3221.3  -544.7     3.9   625.4  1760.1
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      1052.1      142.1   7.406 9.38e-13 ***
## temp                             5290.6      263.7  20.063  < 2e-16 ***
## weathersitMist/Cloudy            -733.7      254.6  -2.882  0.00419 **
## weathersitLight Rain/Snow       -1279.5      805.0  -1.589  0.11283
## temp:weathersitMist/Cloudy        618.0      499.2   1.238  0.21648
## temp:weathersitLight Rain/Snow   -976.5     1755.4  -0.556  0.57837
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 799.9 on 359 degrees of freedom
## Multiple R-squared:  0.668,  Adjusted R-squared:  0.6634
## F-statistic: 144.5 on 5 and 359 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.67, which is an improvement over the temp-only model. The interaction terms are not significant, suggesting that the effect of temperature on rentals does not differ significantly across weather situations. However, the main effect of temp and cloudy weather remain strong and significant.

cnt ~ temp*season

```r
model_temp_season <- lm(cnt ~ temp * season, data = bike_2011)
summary(model_temp_season)
```

```
##
## Call:
## lm(formula = cnt ~ temp * season, data = bike_2011)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3344.1  -307.4   120.4   438.9  1922.7
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)          638.3      237.1   2.691  0.00745 **
## temp                3759.8      818.0   4.596 5.97e-06 ***
## seasonSummer        -201.2      388.7  -0.518  0.60507
## seasonFall          3652.7      783.0   4.665 4.37e-06 ***
## seasonWinter        1349.2      401.4   3.361  0.00086 ***
## temp:seasonSummer   2488.4      990.0   2.513  0.01240 *
## temp:seasonFall    -3512.5     1337.8  -2.626  0.00902 **
## temp:seasonWinter    172.8     1100.6   0.157  0.87531
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 745.7 on 357 degrees of freedom
```

```
## Multiple R-squared:  0.7131, Adjusted R-squared:  0.7075
## F-statistic: 126.8 on 7 and 357 DF,  p-value: < 2.2e-16
```

This model has an R2 of 0.71, which is better than both the temp-only and season-only models. The interaction terms are significant, indicating that the effect of temperature on rentals varies by season.

cnt ~ month*weathersit

```r
model_month_weathersit <- lm(cnt ~ mnth * weathersit, data = bike_2011)
summary(model_month_weathersit)
```

```
##
## Call:
## lm(formula = cnt ~ mnth * weathersit, data = bike_2011)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2538.00  -340.22    42.93   376.17  1569.71
##
## Coefficients: (5 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       1235.53     150.50   8.209 4.91e-15 ***
## mnth2                              616.58     207.17   2.976 0.003131 **
## mnth3                             1099.94     212.85   5.168 4.08e-07 ***
## mnth4                             2506.76     223.96  11.193  < 2e-16 ***
## mnth5                             3374.94     207.17  16.291  < 2e-16 ***
## mnth6                             3710.33     202.46  18.327  < 2e-16 ***
## mnth7                             3313.54     192.13  17.246  < 2e-16 ***
## mnth8                             3436.95     198.48  17.316  < 2e-16 ***
## mnth9                             3628.65     240.12  15.112  < 2e-16 ***
## mnth10                            3373.30     209.87  16.073  < 2e-16 ***
## mnth11                            2402.02     204.71  11.734  < 2e-16 ***
## mnth12                            1554.87     204.71   7.596 3.11e-13 ***
## weathersitMist/Cloudy               47.47     228.63   0.208 0.835646
## weathersitLight Rain/Snow         -729.53     638.54  -1.143 0.254064
## mnth2:weathersitMist/Cloudy       -452.35     339.60  -1.332 0.183760
## mnth3:weathersitMist/Cloudy       -546.77     327.13  -1.671 0.095574 .
## mnth4:weathersitMist/Cloudy      -1010.89     324.73  -3.113 0.002012 **
## mnth5:weathersitMist/Cloudy       -639.44     323.47  -1.977 0.048880 *
## mnth6:weathersitMist/Cloudy       -587.88     336.74  -1.746 0.081768 .
## mnth7:weathersitMist/Cloudy         32.46     403.49   0.080 0.935938
## mnth8:weathersitMist/Cloudy      -1066.95     342.27  -3.117 0.001984 **
## mnth9:weathersitMist/Cloudy       -701.40     333.69  -2.102 0.036304 *
## mnth10:weathersitMist/Cloudy     -1147.80     334.92  -3.427 0.000686 ***
## mnth11:weathersitMist/Cloudy      -366.88     355.72  -1.031 0.303118
## mnth12:weathersitMist/Cloudy       297.24     338.10   0.879 0.379956
## mnth2:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth3:weathersitLight Rain/Snow   -451.94     789.25  -0.573 0.567288
## mnth4:weathersitLight Rain/Snow  -2217.76     905.71  -2.449 0.014854 *
## mnth5:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth6:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth7:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth8:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth9:weathersitLight Rain/Snow  -1951.99     755.71  -2.583 0.010220 *
```

```
## mnth10:weathersitLight Rain/Snow -2056.97     746.65  -2.755 0.006192 **
## mnth11:weathersitLight Rain/Snow  -845.02     745.21  -1.134 0.257637
## mnth12:weathersitLight Rain/Snow  -411.37     787.10  -0.523 0.601570
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 620.5 on 334 degrees of freedom
## Multiple R-squared:  0.8141, Adjusted R-squared:  0.7974
## F-statistic: 48.76 on 30 and 334 DF,  p-value: < 2.2e-16
```

This model produces the best R2 so far of 0.814. All of the months are significant predictors of rentals, and some interaction terms are also significant, indicating that the effect of month on rentals varies by weather situation. This makes sense as the impact of weather on rentals is likely to be more pronounced in certain months (e.g cloudy weather may have higher effect in summer months when it's usually hot).

4. Using Backward Selection

```r
# we need to drop the empty weathersit level first
bike_2011 <- bike_2011 %>% mutate(weathersit = droplevels(weathersit))

# from part 3, we use the interaction plus all the predicts
full_model <- lm(cnt ~ season + mnth + holiday + weekday + workingday + weathersit + temp + hum +
windspeed + temp:weathersit + temp:season +
mnth:weathersit,
data = bike_2011)

summary(full_model)
```

```
##
## Call:
## lm(formula = cnt ~ season + mnth + holiday + weekday + workingday +
##     weathersit + temp + hum + windspeed + temp:weathersit + temp:season +
##     mnth:weathersit, data = bike_2011)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2229.81  -257.38    27.98   283.85  1438.92
##
## Coefficients: (6 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       1712.88     261.36   6.554 2.28e-10 ***
## seasonSummer                       244.98     389.27   0.629 0.529588
## seasonFall                        7194.09     836.72   8.598 3.83e-16 ***
## seasonWinter                       894.57     421.11   2.124 0.034421 *
## mnth2                              238.54     195.51   1.220 0.223341
## mnth3                              472.45     230.85   2.047 0.041533 *
## mnth4                             1315.23     314.57   4.181 3.76e-05 ***
## mnth5                             1818.31     343.66   5.291 2.28e-07 ***
## mnth6                             1631.06     376.56   4.331 1.99e-05 ***
## mnth7                             1284.43     409.18   3.139 0.001855 **
## mnth8                             1138.89     391.10   2.912 0.003847 **
## mnth9                             1018.04     387.04   2.630 0.008948 **
## mnth10                            1003.44     315.41   3.181 0.001611 **
## mnth11                             411.12     289.54   1.420 0.156616
```

```
## mnth12                            444.12     239.23   1.856 0.064315 .
## holidayYes                       -423.28     176.46  -2.399 0.017029 *
## weekdayMon                         52.30     103.84   0.504 0.614882
## weekdayTue                        120.97     104.07   1.162 0.245933
## weekdayWed                         97.19     103.82   0.936 0.349909
## weekdayThu                         74.96     102.71   0.730 0.466033
## weekdayFri                        171.35     101.95   1.681 0.093816 .
## weekdaySat                        194.61     101.99   1.908 0.057286 .
## workingdayYes                         NA         NA      NA       NA
## weathersitMist/Cloudy             -397.11     271.65  -1.462 0.144779
## weathersitLight Rain/Snow         -152.97     719.61  -0.213 0.831796
## temp                             3149.78     807.53   3.900 0.000117 ***
## hum                             -1299.48     293.76  -4.424 1.34e-05 ***
## windspeed                       -2069.40     424.63  -4.873 1.74e-06 ***
## weathersitMist/Cloudy:temp        1525.35     865.02   1.763 0.078806 .
## weathersitLight Rain/Snow:temp    -493.27    2128.06  -0.232 0.816848
## seasonSummer:temp                  122.00     992.26   0.123 0.902222
## seasonFall:temp                  -9264.91    1364.08  -6.792 5.49e-11 ***
## seasonWinter:temp                 1263.28    1110.50   1.138 0.256159
## mnth2:weathersitMist/Cloudy        -25.82     294.12  -0.088 0.930109
## mnth3:weathersitMist/Cloudy       -278.90     297.67  -0.937 0.349490
## mnth4:weathersitMist/Cloudy       -885.83     353.17  -2.508 0.012636 *
## mnth5:weathersitMist/Cloudy       -604.94     409.45  -1.477 0.140550
## mnth6:weathersitMist/Cloudy      -1101.78     497.61  -2.214 0.027531 *
## mnth7:weathersitMist/Cloudy       -437.44     569.81  -0.768 0.443242
## mnth8:weathersitMist/Cloudy      -1495.65     509.02  -2.938 0.003543 **
## mnth9:weathersitMist/Cloudy      -1421.30     443.23  -3.207 0.001480 **
## mnth10:weathersitMist/Cloudy     -1237.51     355.89  -3.477 0.000577 ***
## mnth11:weathersitMist/Cloudy      -518.14     345.43  -1.500 0.134616
## mnth12:weathersitMist/Cloudy      -144.86     301.10  -0.481 0.630767
## mnth2:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth3:weathersitLight Rain/Snow   -872.57     698.89  -1.249 0.212769
## mnth4:weathersitLight Rain/Snow  -1857.16     891.19  -2.084 0.037971 *
## mnth5:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth6:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth7:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth8:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth9:weathersitLight Rain/Snow  -2155.79    1013.85  -2.126 0.034249 *
## mnth10:weathersitLight Rain/Snow -1771.11     790.36  -2.241 0.025727 *
## mnth11:weathersitLight Rain/Snow -1133.07     766.57  -1.478 0.140378
## mnth12:weathersitLight Rain/Snow -1304.63     786.04  -1.660 0.097959 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 506.1 on 316 degrees of freedom
## Multiple R-squared:  0.883,  Adjusted R-squared:  0.8653
## F-statistic:  49.7 on 48 and 316 DF,  p-value: < 2.2e-16
```

```r
# using AIC for backward selection
library(MASS)
```

```
##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
```

```
##
##     select
```

```r
final_model <- stepAIC(full_model, direction = "backward", trace = TRUE)
```

```
## Start:  AIC=4590.85
## cnt ~ season + mnth + holiday + weekday + workingday + weathersit +
##     temp + hum + windspeed + temp:weathersit + temp:season +
##     mnth:weathersit
##
##
## Step:  AIC=4590.85
## cnt ~ season + mnth + holiday + weekday + weathersit + temp +
##     hum + windspeed + weathersit:temp + season:temp + mnth:weathersit
##
##                   Df Sum of Sq      RSS    AIC
## - weekday          6   1304537 82233622 4584.7
## - weathersit:temp  2    873468 81802553 4590.8
## <none>                          80929085 4590.9
## - holiday          1   1473655 82402740 4595.4
## - mnth:weathersit 17  10372986 91302071 4600.9
## - hum              1   5011629 85940714 4610.8
## - windspeed        1   6082618 87011703 4615.3
## - season:temp      3  16230058 97159143 4651.6
##
## Step:  AIC=4584.69
## cnt ~ season + mnth + holiday + weathersit + temp + hum + windspeed +
##     weathersit:temp + season:temp + mnth:weathersit
##
##                   Df Sum of Sq      RSS    AIC
## - weathersit:temp  2    758304 82991926 4584.0
## <none>                          82233622 4584.7
## - holiday          1   1820253 84053875 4590.7
## - mnth:weathersit 17   9674307 91907929 4591.3
## - hum              1   5708569 87942192 4607.2
## - windspeed        1   6251719 88485341 4609.4
## - season:temp      3  16251429 98485051 4644.5
##
## Step:  AIC=4584.04
## cnt ~ season + mnth + holiday + weathersit + temp + hum + windspeed +
##     season:temp + mnth:weathersit
##
##                   Df Sum of Sq      RSS    AIC
## <none>                          82991926 4584.0
## - holiday          1   1826173 84818099 4590.0
## - mnth:weathersit 17  12185420 95177347 4600.0
## - hum              1   5505992 88497919 4605.5
## - windspeed        1   5945051 88936977 4607.3
## - season:temp      3  16654316 99646242 4644.8
```

```r
summary(final_model)
```

```
##
## Call:
## lm(formula = cnt ~ season + mnth + holiday + weathersit + temp +
```

```
##     hum + windspeed + season:temp + mnth:weathersit, data = bike_2011)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2184.13  -258.79    31.47   269.45  1555.07
##
## Coefficients: (5 not defined because of singularities)
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       1727.05     242.17   7.132 6.51e-12 ***
## seasonSummer                       189.39     380.50   0.498 0.619015
## seasonFall                        7155.34     832.21   8.598 3.52e-16 ***
## seasonWinter                       897.14     403.89   2.221 0.027024 *
## mnth2                              197.61     192.08   1.029 0.304365
## mnth3                              428.22     225.27   1.901 0.058204 .
## mnth4                             1174.76     295.85   3.971 8.83e-05 ***
## mnth5                             1640.91     310.19   5.290 2.26e-07 ***
## mnth6                             1426.04     329.20   4.332 1.97e-05 ***
## mnth7                             1060.78     359.68   2.949 0.003417 **
## mnth8                              923.53     347.29   2.659 0.008221 **
## mnth9                              834.37     349.33   2.389 0.017489 *
## mnth10                             878.20     294.77   2.979 0.003108 **
## mnth11                             341.11     278.89   1.223 0.222175
## mnth12                             412.16     235.82   1.748 0.081453 .
## holidayYes                        -453.19     169.73  -2.670 0.007966 **
## weathersitMist/Cloudy              -25.37     195.26  -0.130 0.896715
## weathersitLight Rain/Snow         -264.24     533.18  -0.496 0.620516
## temp                              3565.30     758.79   4.699 3.88e-06 ***
## hum                              -1321.59     285.05  -4.636 5.15e-06 ***
## windspeed                        -2015.94     418.45  -4.818 2.24e-06 ***
## seasonSummer:temp                  254.44     956.79   0.266 0.790460
## seasonFall:temp                  -9207.68    1354.90  -6.796 5.18e-11 ***
## seasonWinter:temp                 1282.11    1051.40   1.219 0.223569
## mnth2:weathersitMist/Cloudy         47.61     288.21   0.165 0.868883
## mnth3:weathersitMist/Cloudy       -149.89     278.09  -0.539 0.590258
## mnth4:weathersitMist/Cloudy       -512.01     274.76  -1.863 0.063302 .
## mnth5:weathersitMist/Cloudy       -124.64     273.77  -0.455 0.649207
## mnth6:weathersitMist/Cloudy       -466.61     279.28  -1.671 0.095734 .
## mnth7:weathersitMist/Cloudy        307.27     335.59   0.916 0.360551
## mnth8:weathersitMist/Cloudy       -764.03     286.82  -2.664 0.008113 **
## mnth9:weathersitMist/Cloudy       -832.81     283.95  -2.933 0.003597 **
## mnth10:weathersitMist/Cloudy      -875.72     279.56  -3.133 0.001891 **
## mnth11:weathersitMist/Cloudy      -269.58     302.01  -0.893 0.372717
## mnth12:weathersitMist/Cloudy        22.95     281.00   0.082 0.934950
## mnth2:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth3:weathersitLight Rain/Snow   -947.64     654.74  -1.447 0.148766
## mnth4:weathersitLight Rain/Snow  -1816.56     740.79  -2.452 0.014726 *
## mnth5:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth6:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth7:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth8:weathersitLight Rain/Snow        NA         NA      NA       NA
## mnth9:weathersitLight Rain/Snow  -2300.57     622.13  -3.698 0.000255 ***
## mnth10:weathersitLight Rain/Snow -1831.00     610.92  -2.997 0.002936 **
## mnth11:weathersitLight Rain/Snow -1273.14     612.38  -2.079 0.038403 *
## mnth12:weathersitLight Rain/Snow -1465.83     653.65  -2.243 0.025603 *
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 506.1 on 324 degrees of freedom
## Multiple R-squared:  0.8801, Adjusted R-squared:  0.8653
## F-statistic: 59.43 on 40 and 324 DF,  p-value: < 2.2e-16
```

```r
formula(final_model)
```

```
## cnt ~ season + mnth + holiday + weathersit + temp + hum + windspeed +
##     season:temp + mnth:weathersit
```

- So we first start with the full model, we get AIC of 4590.85. Then we first dropped weekday and because removing it, our model improved to 4584.7, that means weekday added noises not signal. So then we dropped temp:weathesit interaction, it then improved to 4584.0. So then we stopped because removing anything else would increasing the AIC. Hence the final model we keep is cnt ~ season + mnth + holiday + weathersit + temp + hum + windspeed + season:temp + mnth:weathersit. So to my undersanding this part is that AIC balances between model fit and complexity. Basically it only keeps a predictor if it earns its place by improving fit enough to justify the added complexity. And so as we see that weekday and temp:weathersit failed the backwward selection test.
- Another thing, the final model explains 88 percent of variance in daily rental as shown R^2 = 0.88. Temperature is the strongest driver of ridership, however, it varies by season. So as we can see in Fall where rental drops sharply as temperatures cool. Huminity, windspeed, and holidays all negatively impact rentals. Bad weather like Mist/CLoudy, light rain/snow is hurting ridership most during peak summer and fall months when they expect it to be good weather.

5.

```r
# preparing data for 2012
bike_2012 <- bike_raw %>% filter(yr == 1) %>% mutate(weathersit = droplevels(weathersit))
```

```r
# predict and print out the metrics
preds <- predict(final_model, newdata = bike_2012)
```

```
## Warning in predict.lm(final_model, newdata = bike_2012): prediction from
## rank-deficient fit; attr(*, "non-estim") has doubtful cases
```

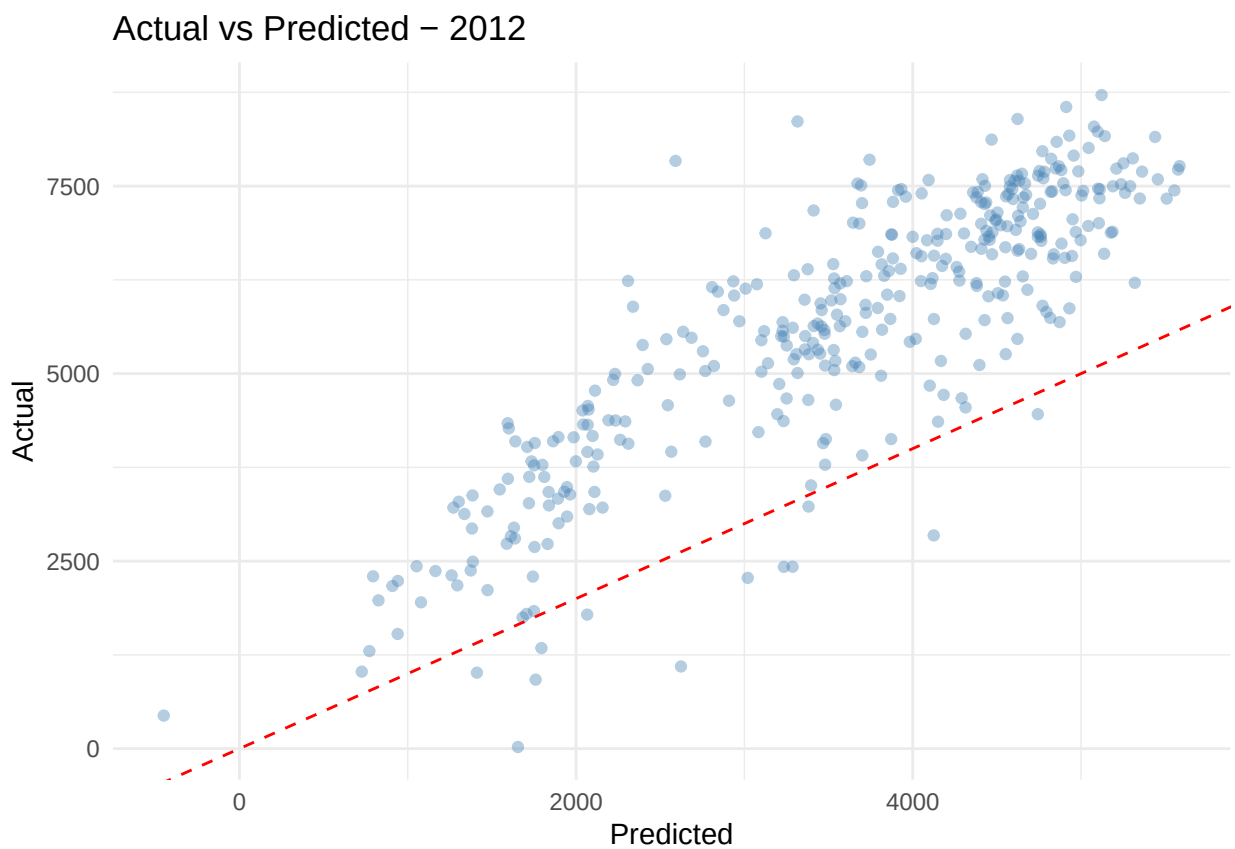
```r
# metrics
rmse <- sqrt(mean((bike_2012$cnt - preds)^2))
r2   <- cor(bike_2012$cnt, preds)^2

rmse
```

```
## [1] 2265.25
```

```r
r2
```

```
## [1] 0.731796
```

-The model performs quite worse comparing to last part. This suggests us that model captured 2011 data pattern well but does not generalize perfectly to 2012. But think about this intuitively, it is likely that ridership grew overall in 2012 as the bike share system matued. So it made sense that a trend the model was not trained to account for.

```r
# visualize the plot
ggplot(bike_2012, aes(x = preds, y = cnt)) + geom_point(alpha = 0.4, color = "steelblue") +
geom_abline(slope = 1, intercept = 0, color = "red", linetype = "dashed") +
```

```r
labs(title = "Actual vs Predicted - 2012", x = "Predicted", y = "Actual") +
theme_minimal()
```